Analysing Severe Adverse Effects

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Contents

{R} # mem.maxVSize(30*1024^3) # plot_model(glmm2, type = "pred", terms = c("age [25:75]", "time", "group"), ci.lvl = 0.95) + # labs(title = "", # y = "Predicted Probability", # x = "Age", # color = "Time") + # scale_y_continuous(labels = function(x) formatC(x, digits = 3, format = "g")) + # theme_classic() # ggsave("figures_tables/pr height = 4, width =8) # 13

data wrangling

```
sae_df <- read_xlsx("data/Q2b.xlsx") |>
  janitor::clean_names() |>
  mutate(sae = ifelse(sae == 1, "Yes", "No") |> as.factor(),
         group = ifelse(group == 1, "Vaccine", "Control") |> as.factor())
#time = as.factor(time),


bl_df <- read_xlsx("data/Q2b_BL.xlsx") |>
  janitor::clean_names() |>
  mutate(sex = ifelse(sex == 1, "Male", "Female"),
         sex = fct_relevel(sex, "Male"))

combined_dataset <- sae_df |>
  full_join(bl_df, by = "id")

combined_dataset_wide <- combined_dataset |>
  pivot_wider(names_from = "time",
              values_from = "sae",
              names_prefix = "month_")
```

Summary Table

```
table1  <- combined_dataset_wide %>%
  dplyr::select(group:age) %>%
  tbl_summary(
    by = group,
    statistic = list(
      all_continuous()  ~ "{mean} ({sd})",    # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)"
```

```
    ),
    label = list(
      #time ~ 'Time (months)',
      age ~ "Age (years)",
      sex ~ "Sex",
      #sae ~ "Severe Adverse Events (Yes)",
      site ~ "Number of Sites")
  ) %>%
  add_p() %>%
  add_overall() %>%
  bold_labels() %>%
  modify_caption("**Baseline Characteristics by Treatment Group**")
# table1v2 <- table1 %>%
#   as_gt() %>%
#   gt::tab_options(
#     data_row.padding = px(2),
#     table.width      = pct(100),
#     table.border.top.color = "black",
#     table.border.bottom.color = "black",
#     heading.border.bottom.color = "black",
#     table.font.color = "black",
#     column_labels.border.bottom.style = "none", # Remove header line
#     table_body.border.bottom.style = "none",    # Remove body line
#     table_body.border.top.style = "none",
#     table.border.top.style = "none",
#     table.font.names = "arial",
#     table.font.size = "12px"
#   ) |>
#   cols_width(
#     label      ~ px(150),
#     everything() ~ px(80)
#   ) |>
#   gt::tab_style(
#     style = gt::cell_borders(),
#     locations = gt::cells_body(columns = everything()))  |>
#   gt::gtsave(filename = "figures_tables/summarytab.pdf")
#system("pdfcrop figures_tables/summarytab.pdf figures_tables/summarytab.pdf")

table1 |>
  as_gt() |>
  cols_width(
    label      ~ px(150),
    everything() ~ px(80)
  ) |>
  gt::gtsave(filename = "figures_tables/summarytab.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp2z0Nd7/file976629bcda57.html screensho
```

```
table1 |>
  as_kable()
```

Table 1: **Baseline Characteristics by Treatment Group**

Characteristic	**Overall** N = 41,250	**Control** N = 20,625	**Vaccine** N = 20,625	**p-value**
Number of Sites	50 (29)	50 (29)	50 (29)	>0.9
Sex				0.2
Male	20,747 (50%)	10,312 (50%)	10,435 (51%)	
Female	20,503 (50%)	10,313 (50%)	10,190 (49%)	
Age (years)	45 (11)	45 (10)	45 (12)	0.3

```
pdf("figures_tables/age_range.pdf", width = 6, height = 3)
ggplot(aes(x = age, fill = group), data = combined_dataset_wide) +
  geom_density(alpha = 0.4) +
  labs(x = "Age", y = "Density", fill = "Tx group") +
  theme_classic()
dev.off()
```

```
## pdf
##   2
```

Checking for missing data pattern (if any)

```
fig2a <- vis_miss(combined_dataset_wide) +
  theme(axis.text.x = element_text(vjust = 0))
ggsave("figures_tables/missingdata_info.png", fig2a, height = 5, width =8)
```

Missingness only seen in outcome data (SAE) and more so for 2nd and third timepoints

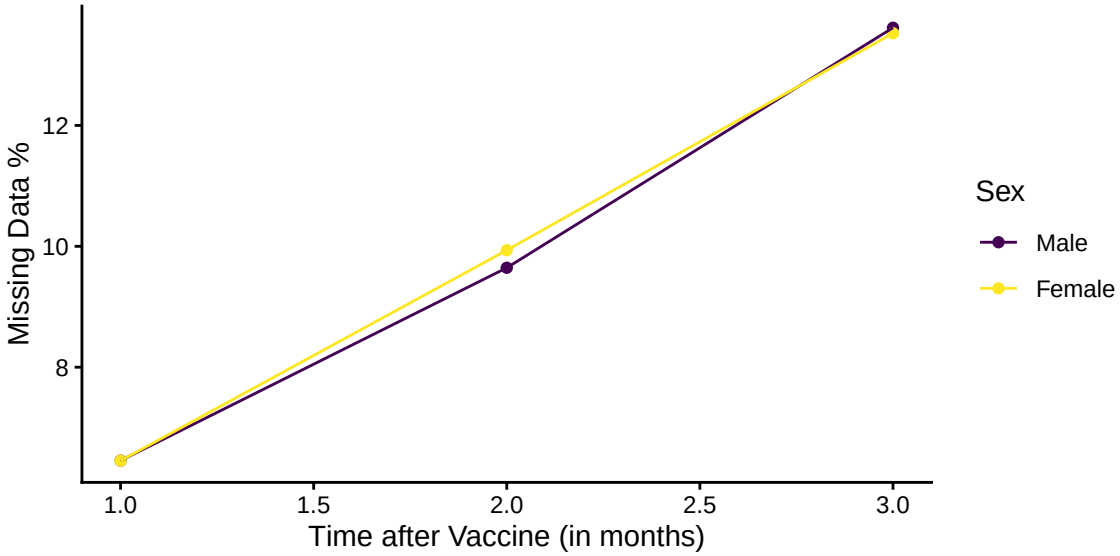
Missingness by gender

```
tx_gender_missing <- combined_dataset |>
  group_by(sex, time) |>
  mutate(total = length(unique(id))) |>
  ungroup() |>
  drop_na() |>
  group_by(sex, time) |>
  summarize (missing_pct = (total[1] - length(unique(id)))*100/total[1])
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by sex and time.
## i Output is grouped by sex.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(sex, time))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
missing_by_gender <- ggplot(aes(x = time, y = missing_pct, group = sex, color = sex),
       data = tx_gender_missing) +
  geom_point() +
  scale_x_continuous(limits = c(1, 3))+
  geom_line() +
  labs(x = "Time after Vaccine (in months)", y = "Missing Data %", color = "Sex") +
```

```
  theme_classic()

ggsave("figures_tables/missingness_by_gender.pdf", missing_by_gender, height = 3, width =5)
#system("pdfcrop figures_tables/missingness_by_gender.pdf figures_tables/missingness_by_gender.pdf")
missing_by_gender
```



Characterise missingness pattern

Number of drop outs

```
missingpattern <- combined_dataset_wide|>
  mutate(pattern = case_when(is.na(month_3) == FALSE & is.na(month_2) == FALSE &
                               is.na(month_1) == FALSE ~ "Complete", ##all or at least 2 where one must
                             is.na(month_3) == FALSE & is.na(month_2) == TRUE &
                               is.na(month_1) == FALSE ~ "Complete",
                             is.na(month_3) == FALSE & is.na(month_2) == FALSE &
                               is.na(month_1) == TRUE ~ "Complete",
                             is.na(month_3) == TRUE & is.na(month_2) == TRUE &
                               is.na(month_1) == TRUE ~ "Assessment Dropout",
                             is.na(month_3) == TRUE & is.na(month_2) == TRUE &
                               is.na(month_1) == FALSE ~ "Early Dropout",
                             is.na(month_3) == TRUE & is.na(month_2) == FALSE &
                               is.na(month_1) == TRUE ~ "Only 2nd Visit",
                             is.na(month_3) == FALSE & is.na(month_2) == TRUE &
                               is.na(month_1) == TRUE ~ "Only 3rd Visit",
                             is.na(month_3) == TRUE ~ "Late Dropout"),
         pattern = fct_relevel(pattern, "Assessment Dropout", "Early Dropout",
                               "Late Dropout"))
missingpattern1 <- missingpattern |>
  dplyr::select(c(sex, pattern)) |>
  tbl_summary(
    by = pattern,
    statistic = list(
      all_continuous()  ~ "{mean} ({sd})",    # drop min/max to save space
```

```
      all_categorical() ~ "{n} ({p}%)"
    ),
    label = list(
      sex ~ "Sex")
  ) %>%
  add_p() %>%
  add_overall() %>%
  bold_labels()

missingpattern1 |>
  as_gt() |>
  cols_width(
    label      ~ px(76),
    everything() ~ px(64)
  ) |>
  tab_options(
    table.font.size = px(10),
    heading.title.font.size = px(11),
    column_labels.font.size = px(10),
    table.font.names = "Arial"
  ) |>
  gt::gtsave(filename = "figures_tables/missingpattern1.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp2z0Nd7/file97662a701eb8.html screensho
```

```
missingpattern1 |>
  as_kable()
```

Characteristic	Overall N = 41,250	Assessment Dropout N = 56	Early Dropout N = 605	Late Dropout N = 4,552	Complete N = 35,364	Only 2nd Visit N = 385	Only 3rd Visit N = 288	p-value
Sex								0.5
Male	20,747 (50%)	25 (45%)	288 (48%)	2,311 (51%)	17,770 (50%)	201 (52%)	152 (53%)	
Female	20,503 (50%)	31 (55%)	317 (52%)	2,241 (49%)	17,594 (50%)	184 (48%)	136 (47%)	

```
missingpatternv2 <- missingpattern |>
  dplyr::select(c(group, pattern)) |>
  tbl_summary(
    by = group,
    statistic = list(
      all_continuous()  ~ "{mean} ({sd})",    # drop min/max to save space
      all_categorical() ~ "{n} ({p}%)"
    ),
    label = list(
      pattern ~ "Missingness Pattern")
  ) %>%
  add_p() %>%
  add_overall() %>%
  bold_labels()
```

```
missingpatternv2 |>
  as_gt() |>
  gt::gtsave(filename = "figures_tables/missingpatternv2.pdf")
```

```
## file:////var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//Rtmp2z0Nd7/file97666346a69c.html screensho
```

```
missingpatternv2 |>
  as_kable()
```

Characteristic	**Overall** N = 41,250	**Control** N = 20,625	**Vaccine** N = 20,625	**p-value**
Missingness Pattern				<0.001
Assessment Dropout	56 (0.1%)	2 (<0.1%)	54 (0.3%)	
Early Dropout	605 (1.5%)	59 (0.3%)	546 (2.6%)	
Late Dropout	4,552 (11%)	1,362 (6.6%)	3,190 (15%)	
Complete	35,364 (86%)	19,126 (93%)	16,238 (79%)	
Only 2nd Visit	385 (0.9%)	43 (0.2%)	342 (1.7%)	
Only 3rd Visit	288 (0.7%)	33 (0.2%)	255 (1.2%)	

Only 3.7% missingness, ignoring for now

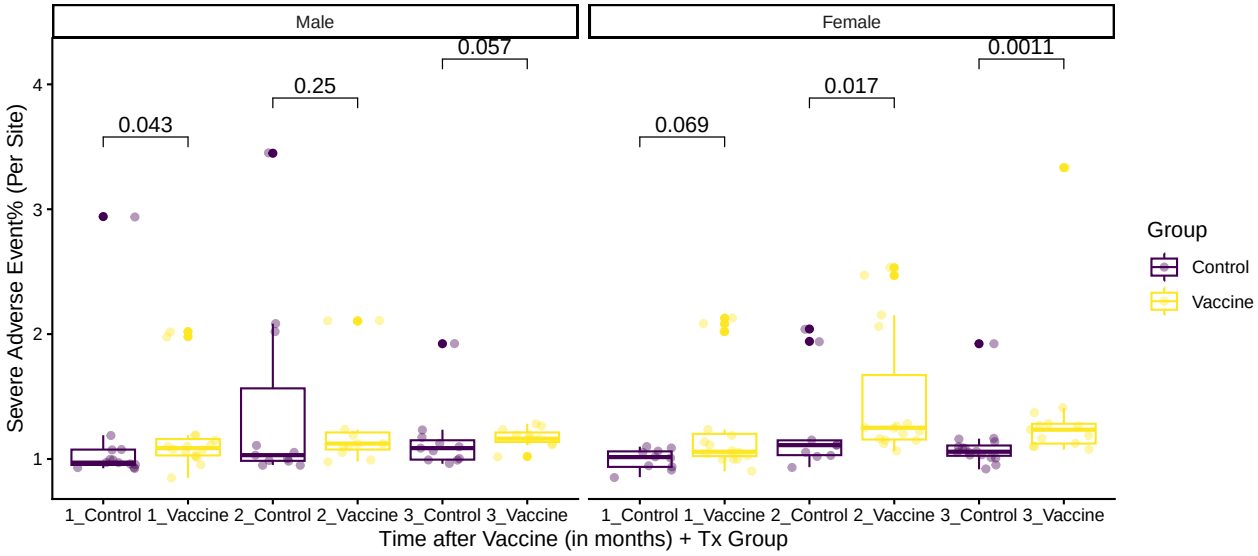
EDA for SAE

```
sae_pct <- combined_dataset |>
  drop_na() |>
  group_by(group, time, site, sex) |>
  mutate(total = n()) |>
  ungroup() |>
  dplyr::filter(sae == "Yes") |>
  group_by(group, time, site, sex) |>
  summarise (sae_pct = n()*100/total[1]) |>
  mutate(group_time = paste(time, group, sep = "_"))
```

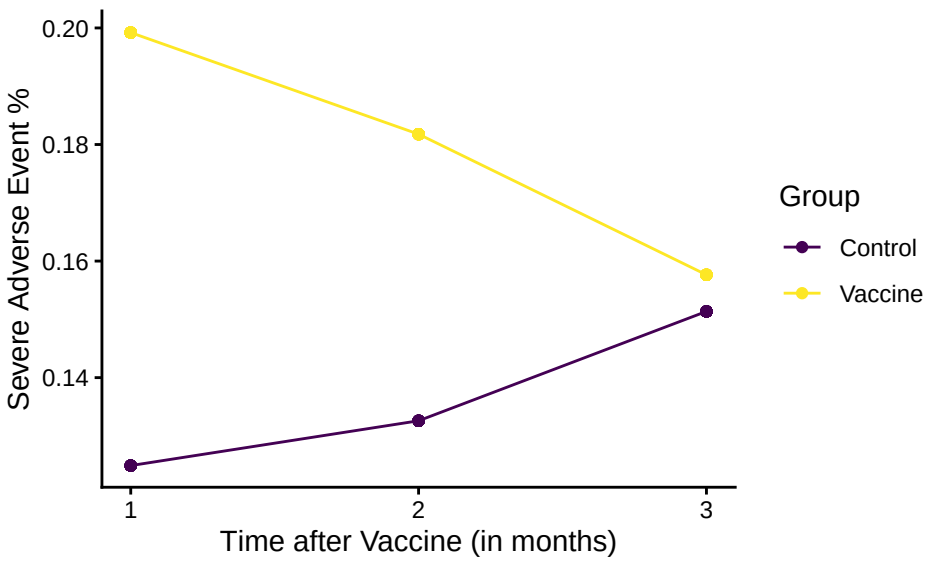
```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by group, time, site, and sex.
## i Output is grouped by group, time, and site.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(group, time, site, sex))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
sae_pct_by_gender_persite <- ggplot(aes(x = group_time, y = sae_pct, color = group),
       data = sae_pct) +
  geom_boxplot() +
  geom_jitter(alpha = 0.4) +
  stat_compare_means(comparisons = list(c("1_Control", "1_Vaccine"),
                                        c("2_Control", "2_Vaccine"),
                                        c("3_Control", "3_Vaccine")),
                     label = "p.adj.format",
                     p.adjust.method = "bonferroni") +
  labs(x = "Time after Vaccine (in months) + Tx Group", y = "Severe Adverse Event% (Per Site)", color =
```

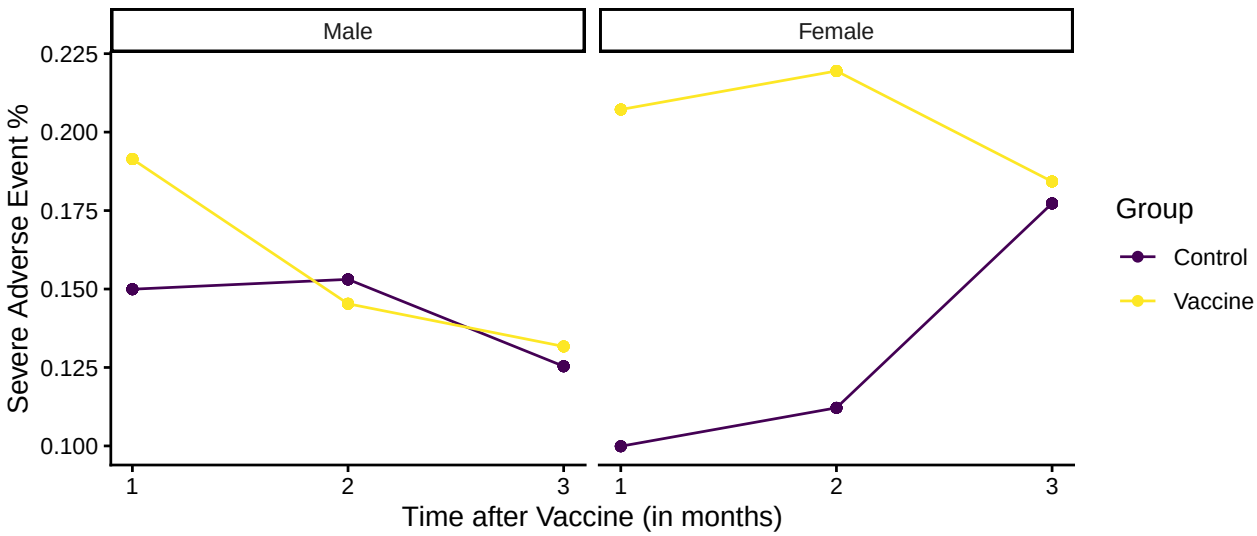
```
  facet_wrap(~sex) +
  theme_classic()

ggsave("figures_tables/sae_pct_by_gender_persite.pdf", sae_pct_by_gender_persite, height = 5, width =11
#system("pdfcrop figures_tables/sae_pct_by_gender_persite.pdf figures_tables/sae_pct_by_gender_persite.
sae_pct_by_gender_persite
```




```
sae_pct <- combined_dataset |>
  drop_na() |>
  group_by(group, time) |>
  mutate(total = n()) |>
  ungroup() |>
  dplyr::filter(sae == "Yes") |>
  group_by(group, time) |>
  mutate (sae_pct = n()*100/total[1])

sae_pct_by_gender <- ggplot(aes(x = time, y = sae_pct, group = group, color = group),
       data = sae_pct) +
  geom_point() +
  scale_x_continuous(limits = c(1, 3), breaks = c(1, 2, 3))+
  geom_line() +
  labs(x = "Time after Vaccine (in months)", y = "Severe Adverse Event %", color = "Group") +
  theme_classic()

ggsave("figures_tables/sae_pct_by_gender.pdf", sae_pct_by_gender, height = 3, width =5)
#system("pdfcrop figures_tables/sae_pct_by_gender.pdf figures_tables/sae_pct_by_gender.pdf")
sae_pct_by_gender
```

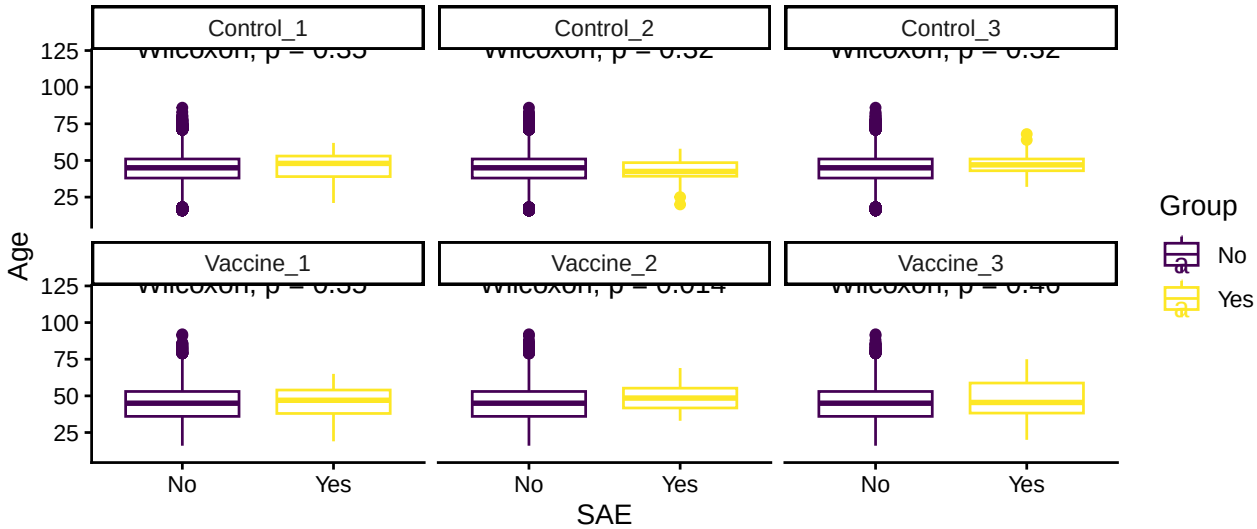



```
sae_pct_by_group <- combined_dataset |>
  drop_na() |>
  group_by(sex, time, group) |>
  mutate(total = n()) |>
  ungroup() |>
  dplyr::filter(sae == "Yes") |>
  group_by(sex, time, group) |>
  mutate (sae_pct = n()*100/total[1])

sae_pct_by_gender_group <- ggplot(aes(x = time, y = sae_pct, group = group, color = group),
       data = sae_pct_by_group) +
  geom_point() +
  scale_x_continuous(limits = c(1, 3), breaks = c(1, 2, 3))+
  geom_line() +
  facet_wrap(~sex) +
  labs(x = "Time after Vaccine (in months)", y = "Severe Adverse Event %", color = "Group") +
  theme_classic()

ggsave("figures_tables/sae_pct_by_gender_group.pdf", sae_pct_by_gender_group, height = 3, width =7)
#system("pdfcrop figures_tables/sae_pct_by_gender_group.pdf figures_tables/sae_pct_by_gender_group.pdf"
sae_pct_by_gender_group
```



###sae by age per group

```
sae_by_age <- ggplot(aes(x = sae, y = age, color = sae),
       data = combined_dataset |> drop_na(sae) |>
  mutate(group_time = paste(group, time, sep = "_"))) +
  geom_boxplot() +
  labs(x = "SAE", y = "Age", color = "Group") +
  facet_wrap(~group_time, ncol = 3)+
  scale_y_continuous(limits = c(10, 120)) +
  stat_compare_means() +
  theme_classic()

ggsave("figures_tables/sae_by_age.pdf", sae_by_age, height = 6, width =8)
#system("pdfcrop figures_tables/sae_pct_by_gender_group.pdf figures_tables/sae_pct_by_gender_group.pdf"
sae_by_age
```



Interaction term not needed read on correlation structure, addin two separate random intercept etc.

PMM: run GLMM per pattern

```r
long_df_w_pattern <- missingpattern |>
  pivot_longer(cols = month_1:month_3,
               names_to = "time",
               values_to = "sae") |>
  mutate(time = as.character(time))

pattern_list <- setdiff(na.omit(as.character(unique(long_df_w_pattern$pattern))),
                        "Assessment Dropout")
run_glmm <- lapply(pattern_list, function(pat){
  df <- long_df_w_pattern |> dplyr::filter(pattern == pat) |>
    mutate(sae = as.character(sae),
           time = as.character(time)) |>
    drop_na()
  if (length(unique(na.omit(df$sae))) > 1 & length(unique(df$time)) > 1){
    print(pat)
    print(unique(df$time))
    df <- df |>
      mutate(sae = as.factor(sae),
             time = as.factor(time))
    fit_PMM <- glmer(sae ~ group + time + sex + age + (1 | site),
                     data = df, family = binomial, nAGQ = 0)
  }
  if (length(unique(na.omit(df$sae))) > 1 & length(unique(df$time)) == 1){
    print(pat)
    print(unique(df$time))
    df <- df |>
      mutate(sae = as.factor(sae))
    fit_PMM <- glmer(sae ~ group + sex + age + (1 | site),
                     data = df, family = binomial, nAGQ = 0)
  }
  if (length(unique(na.omit(df$sae))) == 1 & length(unique(df$time)) == 1){
    fit_PMM <- NULL
  }
  return(fit_PMM)
})
```

```
## [1] "Complete"
## [1] "month_1" "month_2" "month_3"
## [1] "Late Dropout"
## [1] "month_1" "month_2"

## boundary (singular) fit: see help('isSingular')

## [1] "Only 2nd Visit"
## [1] "month_2"
```

```r
names(run_glmm) <- pattern_list
run_glmm <- Filter(Negate(is.null), run_glmm)
```

##Compare coefficients

```r
pairs <- combn(names(run_glmm), 2, simplify = FALSE)

lapply(pairs, function(compare){
  first_fit <- run_glmm[[compare[1]]]
  second_fit <- run_glmm[[compare[2]]]
  coef_first <- fixef(first_fit)
  coef_second <- fixef(second_fit)

  vcov_first <- vcov(first_fit)
  vcov_second <- vcov(second_fit)

  overlapping_terms <- intersect(names(coef_first),
                                 names(coef_second))
  res_complete_late <- lapply(overlapping_terms, function(term){

    beta1 <- coef_first[term]
    beta2 <- coef_second[term]

    var1 <- vcov_first[term, term]
    var2 <- vcov_second[term, term]

    z <- (beta1 - beta2) / sqrt(var1 + var2)
    p <- 2 * (1 - pnorm(abs(z)))

    data.frame(
      `Group` = paste0(compare[1], " vs ", compare[2]),
      `Beta1` = beta1, `Beta2` = beta2, `z` = z, `P-value` = p
    )
  })
  comp <- knitr::kable(as.data.frame(do.call(rbind, res_complete_late)), digits = 3) |>
    kable_styling(latex_options = c("striped", "hold_position"), font_size = 12)
  print(comp)
  #save_kable(comp, file = paste0("figures_tables/coef_comparison", compare[1], compare[2], ".pdf"))
})
```

```
## \begingroup\fontsize{12}{14}\selectfont
##
## \begin{longtable}[t]{llrrrr}
## \toprule
##  & Group & Beta1 & Beta2 & z & P.value\\
## \midrule
## \cellcolor{gray!10}{(Intercept)} & \cellcolor{gray!10}{Complete vs Late Dropout} & \cellcolor{gray!1
## groupVaccine & Complete vs Late Dropout & 0.392 & -0.869 & 2.395 & 0.017\\
## \cellcolor{gray!10}{timemonth\_2} & \cellcolor{gray!10}{Complete vs Late Dropout} & \cellcolor{gray!
## sexFemale & Complete vs Late Dropout & 0.092 & 0.018 & 0.142 & 0.887\\
## \cellcolor{gray!10}{age} & \cellcolor{gray!10}{Complete vs Late Dropout} & \cellcolor{gray!10}{0.013
## \bottomrule
## \end{longtable}
## \endgroup{}
## \begingroup\fontsize{12}{14}\selectfont
##
## \begin{longtable}[t]{llrrrr}
## \toprule
```

```
##  & Group & Beta1 & Beta2 & z & P.value\\
## \midrule
## \cellcolor{gray!10}{(Intercept)} & \cellcolor{gray!10}{Complete vs Only 2nd Visit} & \cellcolor{gray
## groupVaccine & Complete vs Only 2nd Visit & 0.392 & 17.404 & -0.002 & 0.999\\
## \cellcolor{gray!10}{sexFemale} & \cellcolor{gray!10}{Complete vs Only 2nd Visit} & \cellcolor{gray!1
## age & Complete vs Only 2nd Visit & 0.013 & 0.047 & -0.374 & 0.708\\
## \bottomrule
## \end{longtable}
## \endgroup{}
## \begingroup\fontsize{12}{14}\selectfont
##
## \begin{longtable}[t]{llrrrr}
## \toprule
##  & Group & Beta1 & Beta2 & z & P.value\\
## \midrule
## \cellcolor{gray!10}{(Intercept)} & \cellcolor{gray!10}{Late Dropout vs Only 2nd Visit} & \cellcolor{g
## groupVaccine & Late Dropout vs Only 2nd Visit & -0.869 & 17.404 & -0.002 & 0.999\\
## \cellcolor{gray!10}{sexFemale} & \cellcolor{gray!10}{Late Dropout vs Only 2nd Visit} & \cellcolor{gra
## age & Late Dropout vs Only 2nd Visit & 0.023 & 0.047 & -0.254 & 0.800\\
## \bottomrule
## \end{longtable}
## \endgroup{}

## [[1]]
## NULL
##
## [[2]]
## NULL
##
## [[3]]
## NULL
```

Modeling: GLMM to model for odds of SAE at each timepoint

```r
combined_dataset2 <- combined_dataset |>
  mutate(time = as.factor(time))
glmm <- glmer(sae ~ group*time + sex + age + (1 | id),
              family = binomial, data = combined_dataset2, nAGQ = 0)
```

Singularity issue when including site +id or site(id)

Final Model:

$$\begin{aligned}\widehat{logit}(P(SAE_{ijk}=1)) = \beta_0 + \beta_1.I(Group_{ijk} = \text{Vaccine}) + \beta_2.I(Month_{ijk}= 2)+ \\ \beta_3.I(Month_{ijk} = 3) + \beta_4.I(Sex_{ijk} = \text{Female}) + \beta_5.Age_{ijk} + b_i, \\ i = (1,2,\ldots,100), j = (1,2,\ldots,41250),\ k = 1,2,3, \\ b_i \sim N(0,{\sigma^2}_b)\end{aligned}$$

Result form GLMM2

```r
glmm_ad_table <- tbl_regression(
  glmm,
  exponentiate = TRUE,
```

Characteristic	OR	95% CI	p-value
(Intercept)	6.21e-04	2.94e-04, 1.31e-03	**<0.001**
Treatment			
Control	—	—	
Vaccine	1.59	0.96, 2.65	0.073
Time (month)			
1	—	—	
2	1.06	0.61, 1.84	0.8
3	1.21	0.71, 2.07	0.5
Sex			
Male	—	—	
Female	1.10	0.82, 1.48	0.5
Age	1.01	1.00, 1.03	**0.035**
Treatment * Time (month)			
Vaccine * 2	0.86	0.42, 1.78	0.7
Vaccine * 3	0.65	0.31, 1.36	0.3

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
  intercept = TRUE,
  label = list(
    group ~ "Treatment",
    time ~ "Time (month)",
    sex ~ "Sex",
    age ~ "Age"),
  estimate_fun = function(x) {
  ifelse(abs(x) < 1e-2,
         formatC(x, format = "e", digits = 2),
         style_number(x, digits = 2))
}) %>%
  bold_p() %>%
  bold_labels()
glmm_ad_table
```

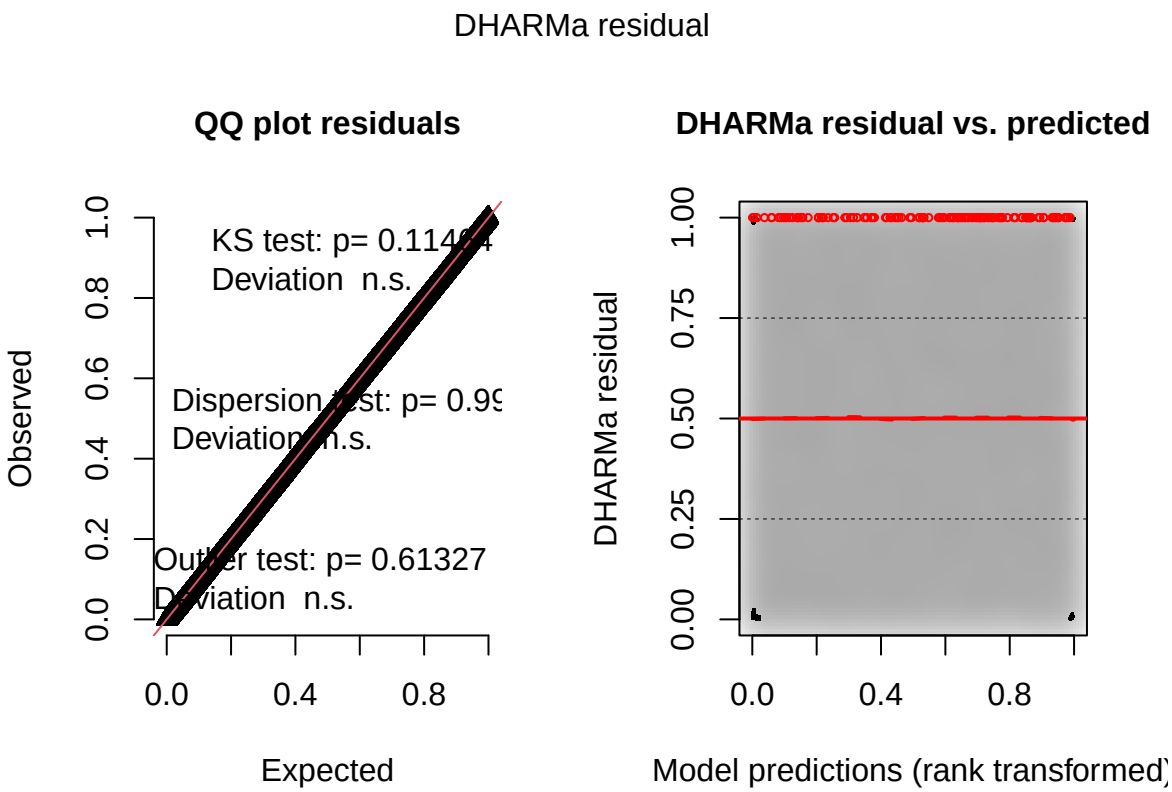
```
{R} # mem.maxVSize(30*1024^3) # plot_model(glmm2, type = "pred",
terms = c("age [25:75]", "time", "group"), ci.lvl = 0.95) +
#   labs(title = "",   #        y = "Predicted Probability", #
x = "Age",  #      color = "Time") + #   scale_y_continuous(labels
= function(x) formatC(x, digits = 3, format = "g")) + #   theme_classic(
# ggsave("figures_tables/predicted_prob.pdf", height = 4, width
=8) #
```

##Diagnostics

```r
sim_ad <- simulateResiduals(fittedModel = glmm)
pdf("figures_tables/model_diganostics.pdf", width = 7, height = 5)
plot(sim_ad)
dev.off()
```

```
## pdf
##   2
```

```r
plot(sim_ad)
```




```r
#check_model(glmm)
# pdf("figures_tables/model_diganostics2.pdf", width = 10, height = 12)
# check_model(glmm)
# dev.off()
```